WANG LABORATORIES, INC.

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DATA SHEET

The Wang 600 Advanced Programmable Calculator combines the easy keyboard operation of manual calculators with the sophistication and power of large programmable calculators, a valuable combination for the scientific and engineering user. The complete set of fully interfaced peripheral equipment, including an output writer, plotters, a paper tape reader, an on-line interface, a magnetic tape cassette drive, plus various other input and output devices, provides the flexibility for designing the calculating system for every application.

The 600 Series Calculators feature sixteen user-definable *special function keys.* In a given program these keys are readily assigned up to 32 specific functions or subroutines which can be executed by a single keystroke. This allows the user to write specific single keystroke programs in each application.

The set of single keystroke technical functions includes the trigonometric, exponential and logarithmic (both to the base e and the base 10) functions. In addition square, square-root, integer, reciprocal and conversion from degrees to radians and from radians to degrees all can be performed as single keystrokes. Floating decimal point notation is used throughout, which is automatically converted to scientific notation when overflow or underflow occurs.

The 600 Series has an optional *Magnetic tape cassette drive* for recording programs and data. Programs can therefore be chained together for automatic execution. The tape drive can also search for a particular program or set of data.

Wang Laboratories provides software and field support enabling the system to be installed and operating on the day it is delivered.

ADVANCED PROGRAMMABLE CALCULATOR 600

The 600 calculating system consists of a calculator and various input and output devices to communicate with the calculator. A standard system consists of the calculator with Panaplex® display, the magnetic tape cassette drive and either the column printer or the fully alphanumeric 601 Output Writer.

The 600 calculator is available in three different memory configurations.

600-2	312 program steps	55 total registers
600-6	824 program steps	119 total registers
600-14	1848 program steps	247 total registers

In each configuration sixteen of the storage registers are dedicated to data only. The remaining registers are part of the program storage area, and are exchangeable with program steps in the storage area. If memory requirements expand, the smaller memories can be retrofitted to the larger capacities.

MAGNETIC TAPE CASSETTE UNIT

The 600 calculator can be ordered with the magnetic tape cassette drive. Programs and data can therefore be recorded for future reference. Since the instruction to LOAD information from tape is programmable, programs can be chained together for automatic execution to a maximum of approximately 30,000 program steps. A selective SEARCH capability enables the calculator to find a specific block of information and load it into memory. Data can also be readily stored on tape under program control for future reference.

COLUMN PRINTER

The optional column printer prints input and output data at a rate of 160 lines per minute. The format of the output is programmable such that fixed or floating point or scientific notation can be designated. Sixteen letters are available for labeling appropriate lines. A programmable TRACE mode uses the printer to print interim answers while a program is being run. In addition a program LISTING can be executed on the printer, whereby the four-digit program codes are sequentially printed out and numbered.

PROGRAMMING

Programs are entered in either the LEARN mode by touching keys in the proper sequence, or in the RUN mode by loading instructions from the magnetic tape cassette.

Storage Registers:

- All storage registers of the 600 are accessible either by DIRECT ADDRESSING or by INDIRECT ADDRESSING. Each of the basic 16 hardwired registers can perform ADD, SUBTRACT, MULTIPLY, DIVIDE, EXCHANGE and TOTAL operations. In addition, the remaining software registers can perform these six operations when addressed INDIRECTLY.

Subroutines:

- 256 separate subroutine codes are available to define subroutines. Subroutines can be nested eight levels deep.

Decision Making:

- Nine different decisions can be made in which the subsequent path of the program is determined by the results of calculations. Loops or iterations can thereby be set up which are performed until certain conditions are satisfied. Unconditional branchings to a specified location of the program is done with a simple SEARCH statement.

Computed Relative GO-TO:

- Instructions are available whereby the results of calculations are used to compute at which program step the program is to continue executing.

Program Debugging:

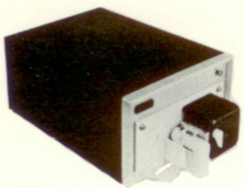
- The TRACE mode can be turned on and off either manually or under program control enabling programs or portions of programs to be analyzed for possible errors. The STEP function causes one step of the program to be executed each time the key is depressed.

Insert and Delete:

- INSERT AND DELETE commands permit the insertions or deletion of a program step with a single keystroke. The SET P.C. (Program Counter) command enables any specific program step to be accessed. Once a step is accessed the STEP and BACKSTEP functions provide single keystroke access to adjacent steps.

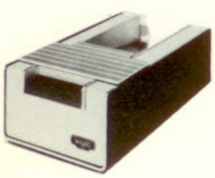
PERIPHERAL EQUIPMENT

601



603

611



614

622

The 601 *Output Writer* types alphabetic and numeric output from the calculator with full format control.*

In the 602 *Plotter,* complete digital plotting is combined with the alphanumeric capability of the 601. Plots are therefore titled and labeled.*

With the 603 *Paper Tape Reader,* raw data is automatically input into the Wang Calculating System providing for an efficient "data reduction" system.

Instrumentation and analytical equipment are interfaced directly to the calculator using the Model 605 *Micro Interface.* "On-line" data acquisition is therefore facilitated.

Large programs and sets of data can be stored easily and inexpensively on the 609 Dual Tape Cassette Drive. Each 150' tape cassette can store a maximum of 90,000 program steps or 11,250 storage registers.

The Model 611 *Input/Output Writer* provides a convenient means for entering alphabetic information into the calculator as well as the print-out capability of the 601 Output Writer.*

The 612 *Flat Bed Plotter* provides continuous line or point plotting of curves and data, as well as full alphanumeric labeling of problems solved on the Wang Calculating System.

With the 614 *Marked Sense Card Reader,* data and programs can be entered directly into the System. Since the cards are prepared "off-line" without tying up the calculator, the System is more efficient.

The 618 *Extended Memory* provides additional storage capacity in increments of 1,024 program steps. Since the 600 Calculator can address multiple peripherals, more than one 618 can be incorporated into one single system.

In the 600 Calculating System, the 622 *Input Keyboard* enables alphabetic, numeric and formatting instructions to be typed directly into the calculator.

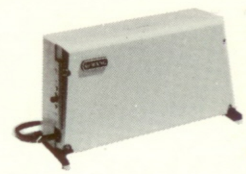
The 629 *Dual Tape Cassette Drive* can store up to 90,000 program steps or 11,250 storage registers. The 629 places "timing marks" on the tape so that the user may "write in place" accurately.

The 621 *High-speed Printer* – will print 132 characters (dot matrix impact printer) per line, at a rate of 150 characters/sec. Up to four carbons may be made.

The 624 *CPU Multiplexer* will allow two 600's to share one external storage device, i.e. a 618, 629, etc.

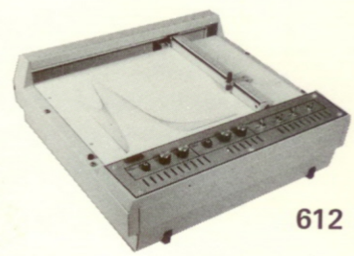
The 627 *Communications Interface* gives the 600 the capability to telecommunicate with another 600 in one of three modes. 1. Core to core, 2. Core to output writer, 3. I/O writer to I/O writer.

602

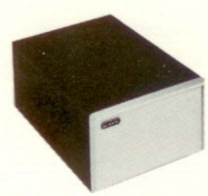


605

609



612



618

*The 601, 602, and 611 can be used as manual, electric typewriters when not being used with the Wang Calculating System.

DATA SHEET

FORMULA PROGRAMMING

The Formula ROM provides an additional set of twenty-five instructions to the 600 such that problems can be solved *algebraically*, in exactly the same sequence as they are written on paper. In addition *one- or two-dimensional subscripting* simplifies the manipulation of the elements of a matrix. The ROM does not interfere with the regular Model 600 keyboard operations or with its ability to use peripheral devices.

STATISTICS ROM

Two statistical options provide single keystroke performance of complex statistical operations. The *Stat ROM* calculates *Mean, Variance* and *Standard Deviation* as well as complete *Linear Regression Analysis* computations. Factorials, Permutations, Combinations plus Normal and Inverse Normal Distributions also can be automatically calculated. The *Advanced* STAT ROM also solves *Numerical Determinants, Simultaneous Equations, Multiple Linear Regression Analysis* and *Matrix Inversions.*

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Special Function Key Assignments for the Formula Programming ROM.

EVALUATION OF DETERMINANT 01	NORM OF DETERMINANT 02	MATRIX INVERSION 03	SIMULTANEOUS EQUATIONS 04	SORTING		MULT. LINEAR REGRESSION		POLYNOMIAL REGRESSION		OPTION FOR RAW SUM OF DATA	RESIDUAL SUM OF DATA
				INPUT	OUTPUT	INPUT	OUTPUT	INPUT	OUTPUT		

Special Function Key Assignments for the Advanced Statistical ROM.

SPECIFICATIONS

MODEL	PROGRAM STEPS	REGISTERS
600-2	312	55
600-6	824	119
600-14	1848	247

Execution Speeds (milliseconds)	
Add/Subtract	1.7
Multiply/Divide	12.0/17.6
$10^x/\log_{10}x$	44.4/35.3
$\sqrt{x}$	99
$e^x/\log_e x$	61.8/47.2
Trig Functions	82 to 255
Read/Write Cycle	5 μ sec
Dynamic Range	10^{-99} to 10^{+99}
Display	10 digits, sign, decimal, 2-digit exponent with sign
Size: Height	9 inches (22.9 cm)
Width	20 inches (50.8 cm)
Depth	20½ inches (52.1 cm)
Weight	39 lbs. (17.7 kg)
Electrical Requirements:	115 or 230 VAC ± 10%
Operating Environment:	59°F to 95°F (15°C to 35°C) 20% to 80% Relative Humidity

ORDERING SPECIFICATIONS

A programmable calculator with capacity of 312 program steps and 55 storage registers (or as specified). Must have sixteen user-definable special function keys for single keystroke access of subroutines. Must provide single keystroke calculation of trigonometric, exponential and logarithmic (to the base e and the base 10) functions with results in floating or scientific notation. The calculator must be able to make decisions based on nine conditions. Must be able to store data on tape for future access and also to use the results of calculations to compute where the program is to continue. Must be able to nest subroutines 8 deep. All storage registers must be accessible directly or indirectly. Must have optional tape cassette drive, column printer with alphabetic labeling, a Formula ROM and two Statistics ROM's. Must be able to control a plotting output writer, flatbed plotter, card readers, paper tape reader, on-line interface and other devices.

Standard Warranty applies.

Wang Laboratories reserves the right to change specifications and price without prior notice.

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